

ECOSYSTEM FRAMEWORK VALIDATION

NeoSmith AI Agent: Automated QA Validation Playbook

Document Type	User Acceptance Testing (UAT) Playbook	Version	2.0.0
Target System	NeoSmith Agentic GraphRAG System	Status	Release Candidate (RC2)

1. Pre-Testing Setup & Architecture Gateway

This playbook serves as a rigorous testing framework to validate the semantic routing, entity parameter extraction, and structural data integrity of the NeoSmith Agentic GraphRAG system prior to demonstration and production deployment.

To honor the system's strict **Anti-Hardcoding Policy**, engineers must utilize the following initialization diagnostic queries within the Neo4j Browser workspace to extract active entity values straight from the baseline dataset. These real-world values should be mapped directly into the respective playbook parameter slots ([PLACEHOLDER]) before running the operational automated tests.

Target Placeholder	Purpose / Context	Extraction Cypher Query
[SAMPLE_ROLE]	Vague/General Job Title	<pre>MATCH (r:Role) RETURN r.title LIMIT 3;</pre>
[SAMPLE_SKILL]	Technical Skill Entity Name	<pre>MATCH (s:Skill) RETURN s.skill_name LIMIT 3;</pre>
[EMP_A] / [EMP_B]	Connected Personnel Pairs	<pre>MATCH (e1:Employee)-[:COLLABORATES_WITH]->(e2:Employee) RETURN e1.name AS EmpA, e2.name AS EmpB LIMIT 1;</pre>
[SUB_TEAM]	Restructuring Scope Target	<pre>MATCH (e:Employee) WHERE e.department IS NOT NULL RETURN DISTINCT e.department AS AvailableDepartments;</pre>

2. Stage 1: Core Functional & Routing Scenarios

Stage 1 focuses on validating the foundational agentic behaviors, ensuring that the model utilizes semantic routing gateways correctly and selects appropriate underlying specialized tools based on executive user inputs.

Scenario 1: Open Positions & Skill Requirements Discovery

USER PROMPT INPUT:

"Show me any open positions we have related to [SAMPLE_ROLE] and list the official technical skills required for them."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- `find_role_by_natural_language`: Invoked first to execute the anti-typo semantic gateway protocol.
Parameter extracted: `index: "role_title_embeddings"`.
- `uc_general_organization_exploration`: Text2Cypher execution to traverse the matched role's structural requirements, as the first tool's post-processing Cypher doesn't natively map full skill arrays.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (op:OpenPosition)-[:FOR_ROLE]->(r:Role)-[:REQUIRES_SKILL]->(s:Skill)
WHERE toLower(r.title) CONTAINS toLower("[SAMPLE_ROLE]")
RETURN op.id AS VacancyID, op.department AS Department, r.title AS OfficialTitle,
collect(s.skill_name) AS RequiredSkills;
```

Scenario 2: Executive-Exempt Individual Contributor (IC) Staff Count

USER PROMPT INPUT:

"Give me a total count of all operational individual contributors across the entire company. Exclude anyone at the Manager, Director, or C-Level."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- `uc_general_organization_exploration`: Direct routing straight into the Text2Cypher engine, identifying the input as an administrative aggregate counting query.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee)-[:HOLDS_ROLE]->(r:Role)
WHERE NOT r.level IN ['Manager', 'Director', 'C-Level', 'C-level']
RETURN count(distinct e) AS IndividualContributorCount;
```

Scenario 3: Skill-Based Talent Sourcing Prioritized by Network Influence

USER PROMPT INPUT:

"Identify all employees who possess proficiency in [SAMPLE_SKILL]. I need them sorted so that the most influential figures within our informal network appear first."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- `find_skill_by_natural_language`: Semantic vector lookup to resolve the imprecise skill entry. Parameter extracted: query string matching `[SAMPLE_SKILL]`.
- **Automated Tool Termination:** The tool's embedded `post_processing_cypher` automatically takes over downstream structure mapping and returns profiles sorted natively by `e.influence_score DESC`.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (s:Skill)<-[:HAS_SKILL]-(e:Employee)-[:HOLDS_ROLE]->(r:Role)
WHERE toLower(s.skill_name) = toLower("[SAMPLE_SKILL]")
RETURN e.name AS Employee, e.influence_score AS Influence, r.title AS Role
ORDER BY e.influence_score DESC LIMIT 10;
```

Scenario 4: Cultural Project Stakeholder Identification

USER PROMPT INPUT:

"Who are our top 5 informal network influencers across the organization? We need to target them for an upcoming cultural project alignment."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- `uc_general_organization_exploration`: Triggered as a baseline exploration task checking non-hierarchical properties across the global employee set.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee)
RETURN e.name AS Employee, e.department AS Department, e.influence_score AS InfluenceScore
ORDER BY e.influence_score DESC LIMIT 5;
```

Scenario 5: Shortest Path ONA Isolation Check

USER PROMPT INPUT:

"Trace the shortest informal collaboration network path between [EMP_A] and [EMP_B] to check how connected our silos are."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- `uc_ona_collaboration_routing`: Maps direct relational traversal. Parameters extracted: `employeeAName: "[EMP_A]", employeeBName: "[EMP_B]"`.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e1:Employee) WHERE toLower(e1.name) CONTAINS toLower("[EMP_A]")
MATCH (e2:Employee) WHERE toLower(e2.name) CONTAINS toLower("[EMP_B]")
MATCH p = shortestPath((e1)-[:COLLABORATES_WITH*..10]-(e2))
RETURN [n IN nodes(p) | n.name + ' (' + n.department + ')'] AS CollaborationPath,
length(p) AS NetworkDistance;
```

Scenario 6: Dynamic Cross-Departmental Reallocation

USER PROMPT INPUT:

"We are restructuring [SUB_TEAM]. Run a cross-departmental talent reallocation strategy to automatically find open slots anywhere in the company, optimizing for minimal training weeks required."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- `uc_talent_circularity_and_reallocation`: Primary execution tool. Parameters extracted: `sourceScope: "[SUB_TEAM]", targetDepartment: ""` (omitted per the mandatory Contextual Mobility Protocol rules to activate global candidate searches).

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee) WHERE toLower(e.department) CONTAINS toLower("[SUB_TEAM]")
MATCH (open:OpenPosition)-[:FOR_ROLE]->(targetRole:Role)-[:REQUIRES_SKILL]->
(targetSkill:Skill)
MATCH (open)-[:REPORTS_TO]->(manager:Employee)
MATCH (e)-[:HAS_SKILL]->(currentSkill:Skill)
OPTIONAL MATCH (currentSkill)-[eq:EQUIVALENT_TO]->(targetSkill)
WHERE currentSkill = targetSkill OR eq IS NOT NULL
RETURN e.name AS Candidate, targetRole.title AS PositionTitle,
coalesce(eq.reskill_time_weeks, 0) AS TrainingWeeksRequired
ORDER BY TrainingWeeksRequired ASC, e.influence_score DESC LIMIT 15;
```

3. Stage 2: Deep Validation & High-Value Scenarios

Stage 2 subjects the NeoSmith agent framework to edge cases, high-density topologies, and advanced organizational network analysis (ONA) scenarios designed to validate deep corporate decision-making structures.

3.1 Secondary Diagnostic Real-Value Extraction Framework

Execute the following advanced diagnostic queries to pull out highly calibrated real-world variables required to feed the Stage 2 validation scenarios.

Target Variable	Operational Selection Context	Diagnostic Extraction Cypher Query
[LOW_EMP_A] / [LOW_EMP_B]	Individual Contributors from distinct divisions who share a valid multi-hop collaborative pathway.	<pre> MATCH (e1:Employee)- [:COLLABORATES_WITH*2..4]-(e2:Employee) MATCH (e1)-[:HOLDS_ROLE]->(r1:Role) MATCH (e2)-[:HOLDS_ROLE]->(r2:Role) WHERE e1.department <> e2.department AND NOT r1.level IN ['Manager','Director','C-Level','C-level'] AND NOT r2.level IN ['Manager','Director','C-Level','C-level'] RETURN e1.name AS EmpA, e1.department AS DeptA, e2.name AS EmpB, e2.department AS DeptB LIMIT 2; </pre>
[FLIGHT_RISK_EMP]	A non-managerial high-influence individual contributor asset ("Quiet Force").	<pre> MATCH (e:Employee)-[:HOLDS_ROLE]-> (r:Role) WHERE NOT r.level IN ['Manager', 'Director', 'C-Level', 'C-level'] RETURN e.name AS QuietForceName, r.title AS Title, e.department AS Department, e.influence_score AS InfluenceScore ORDER BY e.influence_score DESC LIMIT 5; </pre>
[BRIDGE_DEPT_A] / [BRIDGE_DEPT_B]	Verification of standard cross-boundary testing departments.	<pre> MATCH (e:Employee) RETURN DISTINCT e.department AS DeptNames; (Fallback note: Use 'IT' and 'Engineering' if Product is unmapped) </pre>
[FLIGHT_RISK_EMP2]	An individual whose skill record requires explicit, non-zero reskilling weeks to satisfy global open roles.	<pre> MATCH (cs:Skill)-[eq:EQUIVALENT_TO]-> (ts:Skill) WHERE eq.reskill_time_weeks > 0 MATCH (e:Employee)-[:HAS_SKILL]->(cs) MATCH (op:OpenPosition)-[:FOR_ROLE]-> (r:Role)-[:REQUIRES_SKILL]->(ts) RETURN e.name AS Employee, e.department AS FromDept, cs.skill_name AS CurrentSkill, ts.skill_name AS NeededSkill, </pre>

Target Variable	Operational Selection Context	Diagnostic Extraction Cypher Query
		r.title AS TargetRole, eq.reskill_time_weeks AS Weeks LIMIT 3;

Scenario 4 (Expanded): Grassroots ONA Influencer Mapping

USER PROMPT INPUT:

"Identify the top 5 grassroots informal network influencers who operate strictly as individual contributors. Look at the role level property and exclude any nodes where the level is 'Manager', 'Director', or 'C-Level'. They must have high influence scores and actively collaborate across multiple departments."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- uc_general_organization_exploration: Leveraged to parse employee role level attributes, identify cross-departmental graph boundaries, and aggregate weights dynamically via Text2Cypher.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee)-[:HOLDS_ROLE]->(r:Role)
WHERE NOT r.level IN ['Manager', 'Director', 'C-Level', 'C-level']
MATCH (e)-[:COLLABORATES_WITH]-(other:Employee)
WITH e, r, count(DISTINCT other.department) AS CrossDeptCount
WHERE CrossDeptCount > 1
RETURN e.name AS GrassrootsLeader, e.department AS InternalDepartment, r.title AS
OfficialRole, e.influence_score AS InfluenceScore, CrossDeptCount AS DepartmentsBridged
ORDER BY e.influence_score DESC, CrossDeptCount DESC LIMIT 5;
```

Scenario 5 (Expanded): Multi-Hop Cross-Department Paths

USER PROMPT INPUT:

"Trace the multi-hop informal collaboration pathway from [LOW_EMP_A] to [LOW_EMP_B] to evaluate how information flows across boundaries without executive intervention."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- uc_ona_collaboration_routing: Executes path routing. Parameters mapped: employeeAName: "[LOW_EMP_A]", employeeBName: "[LOW_EMP_B]".

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e1:Employee) WHERE toLower(e1.name) CONTAINS toLower("[LOW_EMP_A]")
MATCH (e2:Employee) WHERE toLower(e2.name) CONTAINS toLower("[LOW_EMP_B]")
MATCH p = shortestPath((e1)-[:COLLABORATES_WITH*..10]-(e2))
RETURN [n IN nodes(p) | n.name + ' (' + n.department + ')'] AS GrassrootsPath, length(p)
AS Hops;
```

Scenario 6 (Expanded): Bulk Departmental Reallocation Matrix

USER PROMPT INPUT:

"We are executing a full-scale operational transition for the department named [SUB_TEAM]. Generate a comprehensive talent circularity matrix matching all affected team members to all compatible open positions company-wide. Do not truncate the result to a single candidate; map out all viable placement options sorted by minimal training weeks required."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- uc_talent_circularity_and_reallocation: Parameters extracted: sourceScope: "[SUB_TEAM]", targetDepartment: "". Engaging the Contextual Mobility Protocol forces a global database sweep, outputting un-truncated structural matching arrays.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee) WHERE toLower(e.department) CONTAINS toLower("[SUB_TEAM]")
MATCH (open:OpenPosition)-[:FOR_ROLE]->(targetRole:Role)-[:REQUIRES_SKILL]->
(targetSkill:Skill)
MATCH (open)-[:REPORTS_TO]->(manager:Employee)
MATCH (e)-[:HAS_SKILL]->(currentSkill:Skill)
OPTIONAL MATCH (currentSkill)-[eq:EQUIVALENT_TO]->(targetSkill)
WHERE currentSkill = targetSkill OR eq IS NOT NULL
RETURN e.name AS AffectedEmployee, e.department AS FromDept, targetRole.title AS
AvailableOpenPosition, open.department AS TargetDept, coalesce(eq.reskill_time_weeks, 0)
AS ReskillWeeks
ORDER BY AffectedEmployee ASC, ReskillWeeks ASC;
```


Scenario 7 (Expanded): Structural Silo Detection

USER PROMPT INPUT:

"Identify organizational structural silos by listing the top 10 employees who are most isolated inside the company, meaning they have the lowest number of total collaboration links across the entire network."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- `uc_general_organization_exploration`: Invokes the Text2Cypher baseline topology logic to run structural edge counter aggregations.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee)-[r:COLLABORATES_WITH]->()
WITH e, count(r) AS CollaborationLinks
RETURN e.name AS IsolatedEmployee, e.department AS IsolatedDepartment, CollaborationLinks
AS TotalConnections
ORDER BY CollaborationLinks ASC LIMIT 10;
```

Scenario 8A (Expanded): Bulk Departmental Reallocation Matrix (Training Weeks > 0)

USER PROMPT INPUT:

"We are planning a reallocation for the team named [SUB_TEAM]. Run a talent mobility analysis to find global open positions where team members could fit, specifically highlighting cases that require reskilling weeks."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- uc_talent_circularity_and_reallocation: **Parameters extracted:** sourceScope: "[SUB_TEAM]", targetDepartment: "". Invokes the global cross-functional sweep, isolating elements with functional training distances.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee) WHERE toLower(e.department) CONTAINS toLower("[SUB_TEAM]")
MATCH (open:OpenPosition)-[:FOR_ROLE]->(targetRole:Role)-[:REQUIRES_SKILL]->
(targetSkill:Skill)
MATCH (e)-[:HAS_SKILL]->(currentSkill:Skill)
MATCH (currentSkill)-[eq:EQUIVALENT_TO]->(targetSkill)
WHERE eq.reskill_time_weeks > 0
RETURN e.name AS Candidate, currentSkill.skill_name AS HasSkill, targetSkill.skill_name AS
NeedsSkill, targetRole.title AS Position, eq.reskill_time_weeks AS TrainingWeeks
ORDER BY TrainingWeeks ASC;
```

Scenario 8B (Expanded): Single-Person Targeted Reallocation

USER PROMPT INPUT:

"Execute a targeted talent mobility routing analysis specifically for Jake Peralta. Map his skills against global open positions and check if there are placements available that require reskilling weeks."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- uc_talent_circularity_and_reallocation: **Parameters extracted:** employeeNames: "Jake Peralta", sourceScope: "", targetDepartment: "". Evaluates specialized, non-linear career trajectories based on individual capability matrices.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee) WHERE toLower(e.name) CONTAINS toLower("Jake Peralta")
MATCH (open:OpenPosition)-[:FOR_ROLE]->(targetRole:Role)-[:REQUIRES_SKILL]->
(targetSkill:Skill)
MATCH (e)-[:HAS_SKILL]->(currentSkill:Skill)
OPTIONAL MATCH (currentSkill)-[eq:EQUIVALENT_TO]->(targetSkill)
WHERE currentSkill = targetSkill OR eq IS NOT NULL
RETURN e.name AS Candidate, currentSkill.skill_name AS CurrentSkill,
targetSkill.skill_name AS RequiredSkill, targetRole.title AS PositionTitle,
coalesce(eq.reskill_time_weeks, 0) AS TrainingWeeks
ORDER BY TrainingWeeks DESC LIMIT 10;
```

High-Value Scenario 1: The Critical Flight Risk Vacancy Mitigation (Quiet Forces)

USER PROMPT INPUT:

"If [FLIGHT_RISK_EMP] leaves the company, perform a structural safety analysis. Execute the reallocation tool and ensure you pass an empty string for the source scope parameter. Find high-priority open positions company-wide."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- uc_talent_circularity_and_reallocation: Parameters mapped: employeeNames: "[FLIGHT_RISK_EMP]", sourceScope: "", targetDepartment: "". Explicitly forcing sourceScope as an empty string bypasses parsing structural failures and prevents prompt-chain syntax errors.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee) WHERE toLower(e.name) CONTAINS toLower("[FLIGHT_RISK_EMP]")
MATCH (open:OpenPosition)-[:FOR_ROLE]->(targetRole:Role)-[:REQUIRES_SKILL]->
(targetSkill:Skill)
MATCH (e)-[:HAS_SKILL]->(currentSkill:Skill)
OPTIONAL MATCH (currentSkill)-[eq:EQUIVALENT_TO]->(targetSkill)
WHERE currentSkill = targetSkill OR eq IS NOT NULL
RETURN e.name AS TargetEmployee, targetRole.title AS RecommendedOpenPosition,
open.department AS TargetDepartment, coalesce(eq.reskill_time_weeks, 0) AS
TrainingWeeksRequired
ORDER BY TrainingWeeksRequired ASC LIMIT 10;
```

High-Value Scenario 2: ONA Silo Bridge Analysis

USER PROMPT INPUT:

"Execute a general organization discovery query to find employees bridging different departments. Specifically, list the connections and the relationship weights between individual nodes inside [BRIDGE_DEPT_A] and nodes inside [BRIDGE_DEPT_B]."

EXPECTED AGENT TOOL EXECUTION CHAIN:

- `uc_general_organization_exploration`: Explicitly routes to Text2Cypher. Intentionally re-focusing the prompt towards structural attribute listing and node property evaluation removes LLM reasoning bottlenecks associated with path calculation weights.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e1:Employee)-[c:COLLABORATES_WITH]-(e2:Employee)
WHERE toLower(e1.department) CONTAINS toLower("[BRIDGE_DEPT_A]")
AND toLower(e2.department) CONTAINS toLower("[BRIDGE_DEPT_B]")
RETURN e1.name AS SourceEmployee, e1.department AS SourceDept, e2.name AS TargetEmployee,
e2.department AS TargetDept, c.weight AS BridgeConnectionWeight
ORDER BY c.weight DESC LIMIT 10;
```

High-Value Scenario 3: Hidden Mentor Upskilling Architect

USER PROMPT INPUT:

"We have open positions requiring Python skills. Who are the highly proficient, high-influence Python experts inside the company who could act as upskilling mentors for internal transfers?"

EXPECTED AGENT TOOL EXECUTION CHAIN:

- `find_skill_by_natural_language`: Invoked automatically as part of the Anti-Typo Semantic Gateway to safely translate conversational text into the official corporate 'Python' skill node.
- `uc_general_organization_exploration`: Processes the resulting unique skill identifier token via Text2Cypher to parse and rank subject-matter experts (SMEs) crossing high proficiency and ONA authority levels.

NEO4J DATABASE VALIDATION QUERY:

```
MATCH (e:Employee)-[h:HAS_SKILL]->(s:Skill)
WHERE toLower(s.skill_name) = toLower("Python")
MATCH (e)-[:HOLDS_ROLE]->(r:Role)
RETURN e.name AS Employee, r.title AS Role, h.proficiency AS SkillProficiency,
e.influence_score AS Influence
ORDER BY e.influence_score DESC;
```

4. Appendix: Production Environment Production Test Samples

The following tables contain real, validated node entities populated inside the Neo4j instance. QA engineers can utilize these parameter blocks to substitute playbook placeholders during testing runs.

4.1 Stage 1 Baseline Testing Datasets

[SAMPLE_ROLE] Option	[SAMPLE_SKILL] Option	[EMP_A] / [EMP_B] Pairs	[SUB_TEAM] Option
"Junior Analyst - Engineering"	"Python"	"Thomas Anderson" / "Shannon Campos"	"Finance"
"IT Manager"	"Cybersecurity"	"Morpheus" / "Holly Miller"	"IT"
"Senior Analyst - IT"	"Sales Forecasting"	"Billy Butcher" / "Ashley Rhodes"	"Engineering"
"Finance Specialist"	"Financial Modeling"	"William Sanders" / "Cynthia Palmer"	"People"
N/A	N/A	"Obi-one Kenobi" / "Lindsay Williams"	"Sales" / "Logistics"

4.2 Stage 2 Deep-Traversal Testing Datasets

[LOW_EMP_A] / [LOW_EMP_B] Deep Path Node Pairs	[FLIGHT_RISK_EMP] Target Scopes
"Brian Yates" / "Michael Reed"	"Wanda Swanson"
"Danielle Johnson MD" / "Michael Reed"	"Jeremy Gonzalez"
"Patrick Mcdaniel" / "Robert Freeman"	"Patricia Gonzalez"
"Ernest Carr" / "Robert Freeman"	N/A